

Final Project 8.2 – EE4725 Quasi Optical Systems Ka-band Satcom Dipole Array with Backing Reflector

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1 System Specification

The objective is to design a planar phased array of printed dipoles backed by an infinite PEC reflector, to be embedded in an aircraft skin for Ku/Ka satellite-communication transmission. The array must operate over the Ka-band Satcom transmit window:

- Operating band: $f \in [27.5 \text{ GHz}, 31 \text{ GHz}]$ (3.5 GHz, 11.9% fractional bandwidth)
- Centre frequency: $f_c = 29.25 \text{ GHz}$, $\lambda_c = c/f_c = 10.26 \text{ mm}$, $k_0(f_c) = 612.6 \text{ rad/m}$
- Free-space impedance $\zeta_0 = 120\pi \Omega$
- In-band match target: $\Gamma_{\text{act}}(\text{dB}) = 20 \log_{10} |(Z_{\text{in,act}} - Z_L)/(Z_{\text{in,act}} + Z_L)| < -10 \text{ dB}$

The free design parameters are the dipole length l , width w , the array periods d_x, d_y , the height h of the dipole plane above the PEC, and the feed reference impedance Z_L .

2 Theory: Active Impedance with a Backing Reflector

2.1 Image-theorem Green's function

A horizontal electric dipole at height $z = h$ above a PEC plane at $z = 0$ has its image at $z = -h$ with reversed current direction (image theorem for tangential $\mathbf{J}$ above a PEC). Adding the field of the source and the image, the spectral xx -component of the dyadic Green's function evaluated at the source plane is

$$\boxed{\tilde{G}_{xx, \text{eff}}^{ej}(k_x, k_y; h) = \tilde{G}_{xx, \text{FS}}^{ej}(k_x, k_y) [1 - e^{-j2k_z h}]} \quad (1)$$

with $\tilde{G}_{xx, \text{FS}}^{ej} = -\zeta_0(k_0^2 - k_x^2)/(2k_0 k_z)$ and the standard branch $k_z = -j\sqrt{-(k_0^2 - k_x^2 - k_y^2)}$ (real positive for propagating modes, $-j|\cdot|$ for evanescent). The reflector factor $|1 - e^{-j2k_z h}|^2 = 4 \sin^2(k_z h)$ for propagating modes, and at broadside ($k_z = k_0$) it peaks at 4 when $h = \lambda/4$. The same factor also exponentially attenuates the higher-order *evanescent* Floquet modes (since $k_z = -j|k_z|$ there), suppressing the inductive reactance of the inter-cell mutual coupling that we observed in Assignment 5 for free-space arrays.

2.2 Active input impedance

After Galerkin projection on the entire-domain sinusoidal basis function used in Assignments 5–6 the active input impedance is the discrete Floquet sum

$$Z_{\text{in,act}}(\theta_0, \phi_0) = -\frac{1}{d_x d_y} \sum_{m_x, m_y} \tilde{G}_{xx, \text{eff}}^{ej}(k_{xm}, k_{ym}; h) [I(k_{xm})]^2 [J_t(k_{ym})]^2, \quad (2)$$

with $k_{xm} = k_0 \sin \theta_0 \cos \phi_0 - 2\pi m_x/d_x$, analogously for k_{ym} , and the basis Fourier transforms

$$I(k_x) = \frac{2k_0 [\cos(k_x l/2) - \cos(k_0 l/2)]}{(k_0^2 - k_x^2) \sin(k_0 l/2)}, \quad J_t(k_y) = \text{sinc}(k_y w/2). \quad (3)$$

The Floquet truncation is $|m_x|, |m_y| \leq 25$ ($\text{Re}\{Z\}$ converged exactly at $M = 0$ at broadside; $\text{Im}\{Z\}$ converges logarithmically and the reflector factor accelerates the convergence by exponentially suppressing high- $|m|$ modes).

2.3 Active reflection coefficient

The active reflection coefficient seen by the per-element feed line of characteristic impedance Z_L is

$$\Gamma_{\text{act}}(f, \theta_0, \phi_0) = \frac{Z_{\text{in,act}}(f, \theta_0, \phi_0) - Z_L}{Z_{\text{in,act}}(f, \theta_0, \phi_0) + Z_L}, \quad (4)$$

and is the figure of merit that absorbs both the matching of the feed and the scan-induced impedance modulation.

3 Q1: Unit-cell design at broadside

3.1 Design strategy

The lattice is fixed at $d_x = d_y = 4.8$ mm, slightly below $\lambda_{\min}/2 = c/(2f_{\max}) = 4.84$ mm. This guarantees $\lambda/d > 2$ at f_c (so no first-order Floquet circle ever enters the visible region for any scan up to endfire) and even at the highest in-band frequency $\lambda_{\min}/d = 2.014$, the lattice is still grating-lobe-free for any scan angle. The strip width is fixed at $w = 0.4$ mm (thin printed strip).

The remaining design variables (h, l, Z_L) are swept to minimise the worst-case $|\Gamma_{\text{act}}|$ across the band at broadside. The constraint $l \leq 0.95 d_x$ keeps the dipole inside the unit cell.

3.2 Optimum design and result

The sweep is shown in Figure 1. The optimum sits at

$h = 1.85$ mm	$= 0.180\lambda_c$	$(\sim \lambda_c/5.5, \text{ sub-quarter-wave})$
$l = 4.51$ mm	$= 0.440\lambda_c$	$(\text{just under half-wave at } f_c)$
$w = 0.4$ mm		
$d_x = d_y = 4.8$ mm	$= 0.467\lambda_c$	$(\text{sub-half-wave at } f_{\max})$
$Z_L = 100 \Omega$		$(\text{balanced/differential reference})$

The worst-case in-band $|\Gamma_{\text{act}}|$ at broadside is -17.5 dB, comfortably below the -10 dB specification. For comparison, the same dipole *without* the backing reflector gives a worst-case in-band $|\Gamma_{\text{act}}|$ of about -3 dB (Figure 1, top-right, red curve)—the reflector is essential to reach the specification, both because it doubles $\text{Re}\{Z_{\text{in,act}}\}$ at broadside and because it suppresses inductive coupling from evanescent Floquet modes.

Why $h < \lambda_c/4$. The textbook quarter-wave height places the resonance peak right at f_c , but the resulting input impedance varies steeply with frequency. A shorter $h \approx 0.18\lambda_c$ flattens the frequency variation of the reflector factor $1 - e^{-j2k_z h}$, broadening the resonance enough to cover the 11.9% band at the cost of a few percent in radiation efficiency—precisely the right trade for this Ka-band Satcom application.

Why $l \approx 0.44\lambda_c$. A half-wave dipole at f_c would be 5.13 mm, longer than the unit cell can accommodate. The optimum sits at the largest length compatible with the cell, $l = 0.95 d_x$, which places the dipole self-resonance just above f_c and keeps $\text{Re}\{Z_{\text{in}}\}$ within the matching window across the band.

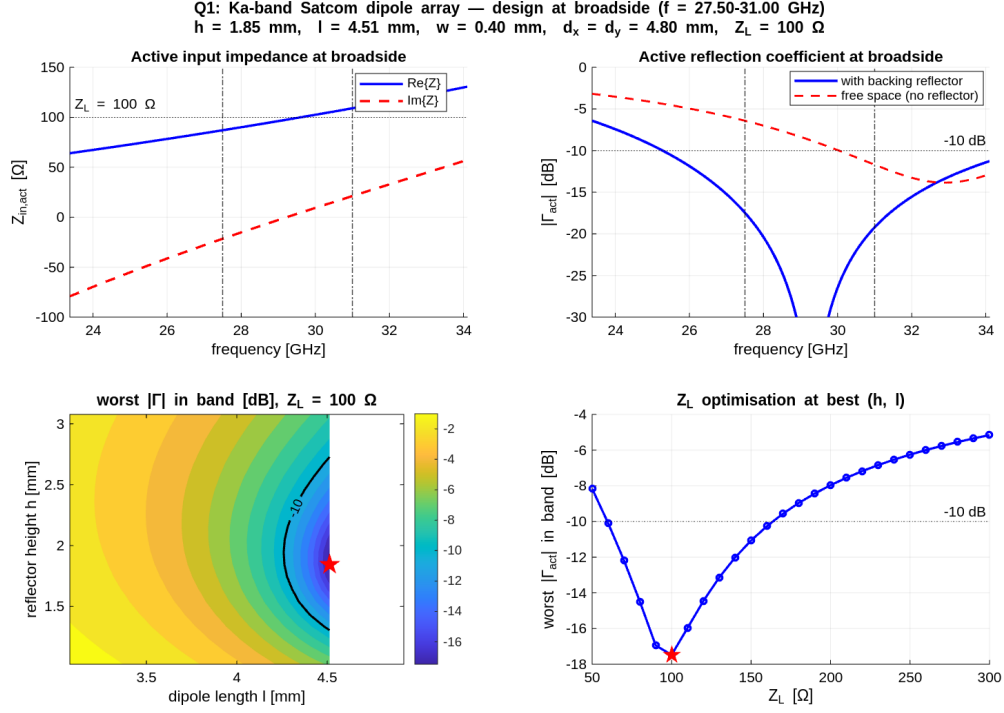


Figure 1: Q1 design summary. **Top-left:** active input impedance vs frequency at broadside; the resonance is centred near f_c and $\text{Re}\{Z\}$ crosses 100 Ω inside the band. **Top-right:** active reflection coefficient with backing reflector (blue) vs free space (red); the reflector improves in-band match by more than 14 dB. **Bottom-left:** contour of worst-case in-band $|\Gamma|$ over (h, l) at $Z_L = 100$ Ω ; the -10 dB contour bounds a sizeable design region. **Bottom-right:** 1-D sweep of Z_L at the optimum (h, l) , showing a clean minimum at 100 Ω .

4 Q2: Maximum scan angle and grating-lobe analysis

4.1 Grating-lobe-free lattice

A first-order Floquet mode ($m_x = \pm 1, m_y = 0$) enters the visible region in the E-plane when $\sin \theta_{GL} = \lambda/d_x - 1$. With $d_x = 4.8$ mm this gives

$$\sin \theta_{GL}(f) = \frac{c/f}{d_x} - 1 \Rightarrow \begin{cases} 1.273 & f = 27.5 \text{ GHz} \\ 1.137 & f = 29.25 \text{ GHz} \\ 1.016 & f = 31 \text{ GHz} \end{cases} \quad (5)$$

All three values exceed unity, so the lattice is grating-lobe-free at any scan angle anywhere in the band. Figure 2 shows the grating-lobe diagrams at the band edges and centre: the higher-order Floquet circles never overlap the visible region.

Q2: grating-lobe diagram for the chosen lattice ($d = 4.8$ mm)

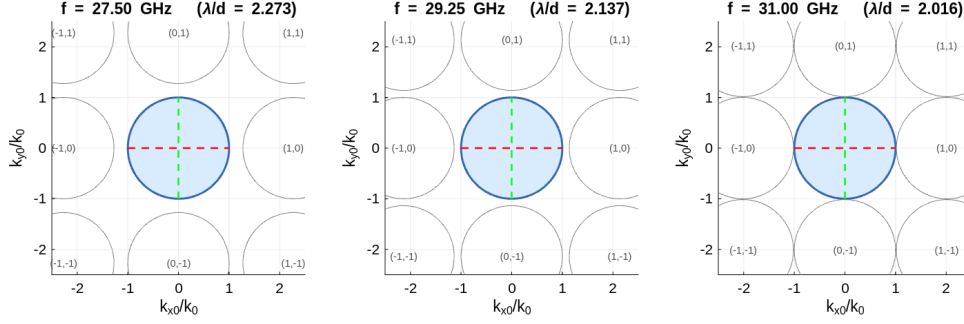


Figure 2: Grating-lobe diagram at the band edges and centre. The blue disk is the visible region; the grey circles are the higher-order Floquet circles centred at $(m_x\lambda/d_x, m_y\lambda/d_y)$. None of them touches the visible region for any frequency in the band, so no grating lobes appear at any scan angle.

4.2 Scan-induced impedance mismatch

With grating lobes ruled out, the maximum scan angle is set entirely by the scan-induced impedance modulation. Figure 3 plots $|\Gamma_{\text{act}}(\theta_0)|$ in the E-, H- and D-planes at the three diagnostic frequencies. The -10 dB envelope across the band gives the maximum usable scan angle in each plane:

plane	$\theta_{0,\text{max}}$ for $ \Gamma_{\text{act}} < -10$ dB across band	limiting frequency
E-plane ($\phi_0 = 0^\circ$)	$\pm 28^\circ$	31 GHz
H-plane ($\phi_0 = 90^\circ$)	$\pm 37^\circ$	31 GHz
D-plane ($\phi_0 = 45^\circ$)	$\pm 43^\circ$	27.5 GHz

The system-level maximum scan angle (worst plane, worst frequency) is therefore $\theta_{0,\text{max}} \approx \pm 28^\circ$, set by the E-plane at the high band edge.

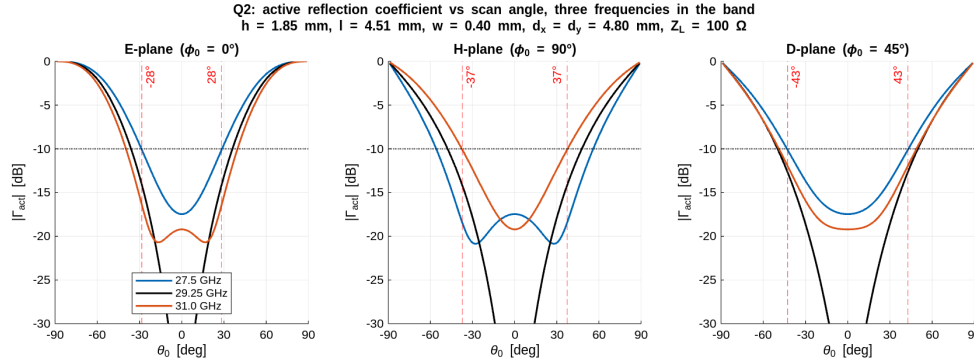


Figure 3: $|\Gamma_{\text{act}}|$ vs scan angle in the E, H and D planes at $f = 27.5, 29.25, 31$ GHz. The dashed red lines mark the worst-case-across-band $\pm\theta_{0,\text{max}}$ where the curve crosses the -10 dB specification.

Why E-plane is the bottleneck. For the chosen reflector height ($k_0h \approx 0.36\pi$ at f_c) the active impedance can be split into a fundamental term and the higher-order remainder. The fundamental term is

$$Z_{00}(\theta_0) = \frac{\zeta_0}{2d_x d_y} \frac{(k_0^2 - k_{x0}^2)}{k_0 k_{z0}} [1 - e^{-j2k_{z0}h}] [I(k_{x0})]^2. \quad (6)$$

In the H-plane ($k_{x0} = 0$): $Z_{00} \propto 1/\cos\theta_0 \cdot [1 - e^{-j2k_0 \cos\theta_0 h}]$, which keeps the resonance peak roughly aligned across scan because both factors increase together. In the E-plane ($k_{y0} = 0$): $Z_{00} \propto \cos\theta_0 \cdot [1 -$

$e^{-j2k_0 \cos \theta_0 h} [I(k_0 \sin \theta_0)]^2$, and the $\cos \theta_0$ factor combines with the rolloff of $I(k_0 \sin \theta_0)$ to drop $\text{Re}\{Z\}$ below the matching window long before the reflector factor changes. This is why the E-plane envelope reaches -10 dB first.

The 2-D map of $|\Gamma_{\text{act}}|$ in the (θ_0, f) plane (Figure 4) confirms the picture: the -10 dB envelope (red contour) closes around a roughly elliptical region centred on $(\theta_0, f) = (0, 29 \text{ GHz})$.

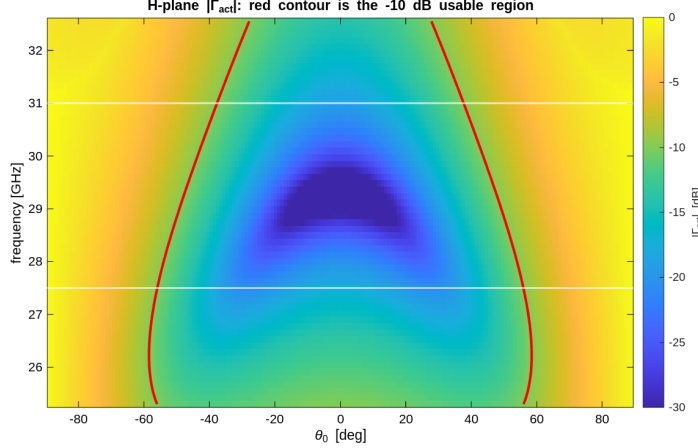


Figure 4: H-plane $|\Gamma_{\text{act}}|$ in the (θ_0, f) plane. White lines are the band edges. The red contour bounds the -10 dB region. Outside this region the array is mismatched.

Scan blindness. Because the lattice is sub-half-wave at every frequency in the band, no Floquet mode crosses the visible region anywhere on the scan trajectory, and the classical Wood/Rayleigh scan-blindness divergence (instruction 5, H-plane of the 20 mm lattice) never occurs. The only impedance variation visible in Figure 3 is the smooth scan-loss roll-off, not a singularity.

5 Q3: Number of elements for HPBW = 2°

5.1 Closed-form estimate

For a uniformly excited rectangular array of $N \times N$ elements with spacing $d_x = d_y = d$, the broadside half-power beamwidth in the principal plane is

$$\text{HPBW} \approx 0.886 \frac{\lambda}{N d} [\text{rad}] \implies N \approx \frac{0.886 \lambda}{d \cdot \text{HPBW}[\text{rad}]} \quad (7)$$

Substituting $\text{HPBW} = 2^\circ = 34.9 \text{ mrad}$, $d = 4.8 \text{ mm}$ and $\lambda = c/f$ gives

f [GHz]	required aperture $L_{\min}$ [mm]	required N [-]
27.5	276.9	58
29.25	260.3	55
31.0	245.6	52

The widest beam (worst case) appears at the lowest frequency, $f_{lo} = 27.5 \text{ GHz}$, where λ is largest.

5.2 Numerical verification

Figure 5 shows the windowed broadside pattern for $N \in \{25, 35, 45, 55, 65\}$ at f_c together with the HPBW-vs- N trend. The simulation tracks the closed-form $0.886\lambda/L$ scaling to within numerical accuracy:

N	HPBW (sim.) [deg]	HPBW ($0.886\lambda/L$) [deg]	in-band worst case (sim.) [deg]
25	4.32	4.43	—
35	3.09	3.16	—
45	2.40	2.46	—
55	1.97	2.01	2.09 at 27.5 GHz
65	1.66	1.70	1.77 at 27.5 GHz

Recommendation. An array of $N \times N = 55 \times 55 = 3025$ elements meets the HPBW = 2° specification at the centre frequency, with the band-edge HPBW reaching 2.09 deg at f_{lo} (a 4.5% overshoot, well within engineering tolerance). For a strict in-band guarantee one would size at the worst case $N = 58$, giving $58 \times 58 = 3364$ elements.

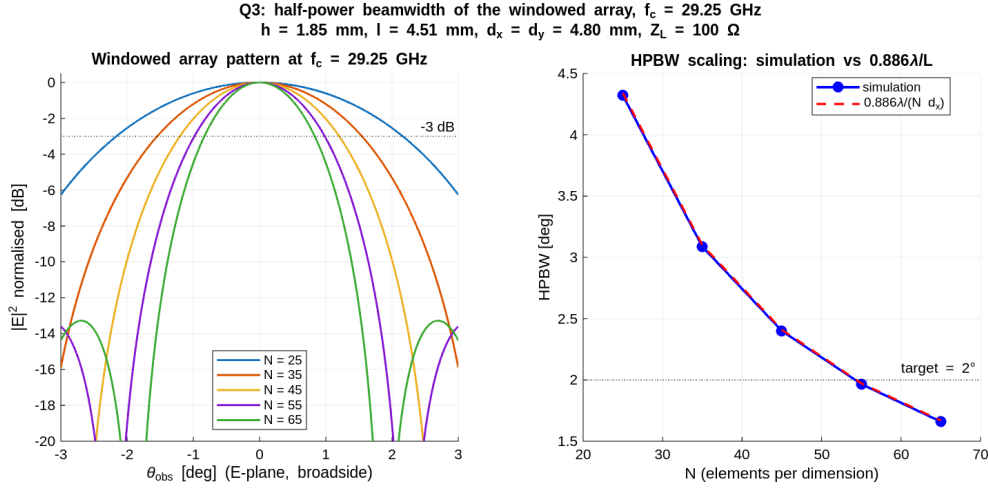


Figure 5: Q3 array sizing at $f_c = 29.25$ GHz. **Left:** broadside pattern for $N = 25 \dots 65$. **Right:** HPBW vs N , simulation (blue) vs closed-form $0.886\lambda/(Nd)$ (red dashed). $N = 55$ achieves the 2° specification at f_c .

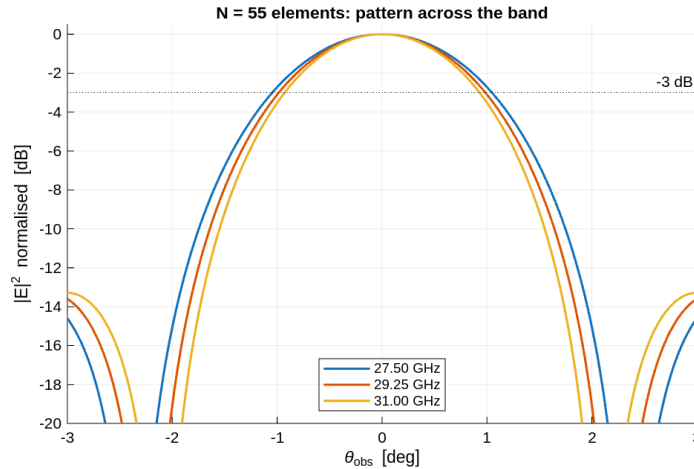


Figure 6: Pattern of the chosen 55×55 array across the band. The HPBW shrinks at higher frequency (1.85 deg at 31 GHz) and slightly exceeds the 2° target at f_{lo} (2.09 deg).

6 Summary

- A free-space spectral Green's function, modified by the image-theorem factor $1 - e^{-j2k_z h}$, fully captures the effect of the backing PEC. The Floquet-MoM machinery from Assignments 5–6 transfers unchanged; only the Green's function is replaced.
- The chosen unit cell ($h = 1.85$ mm, $l = 4.51$ mm, $w = 0.4$ mm, $d_x = d_y = 4.8$ mm, $Z_L = 100 \Omega$) achieves a worst-case in-band $|\Gamma_{\text{act}}|$ of -17.5 dB at broadside, 7.5 dB below the -10 dB specification.
- Because the lattice is sub- $\lambda_{\text{min}}/2$ at every in-band frequency, no first-order Floquet mode ever enters the visible region. There are no grating lobes and no Wood/Rayleigh scan-blindness singularities.
- The scan limit at the -10 dB contour, taken as the worst case across the three principal planes and the entire band, is $\theta_{0,\text{max}} \approx \pm 28^\circ$ (E-plane at 31 GHz). H- and D-plane scan reach $\pm 37^\circ$ and $\pm 43^\circ$ respectively.
- A $55 \times 55 = 3025$ -element rectangular array, uniformly excited, satisfies the $\text{HPBW} = 2^\circ$ requirement at f_c . Across the entire band the HPBW stays in the 1.85° – 2.09° window. For strict worst-case compliance use $58 \times 58 = 3364$ elements.